

PHASE -3

CUSTOMER CHURN PREDICTION

**PREPROCESSING AND LOADING THE DATA
THROUGH THE HELP OF IBM COGNOS.**



Pre Processing



```
1 df = df.drop(['customerID'], axis = 1)
2 df.head()
```

```
1 df['TotalCharges'] = pd.to_numeric(df.TotalCharges, errors='coerce')
2 df.isnull().sum()
```

```
1 df[np.isnan(df['TotalCharges'])]
```

```
1 df[np.isnan(df['TotalCharges'])]
```

```
1 df[df['tenure'] == 0].index
```

```
Int64Index([488, 753, 936, 1082, 1340, 3331, 3826, 4380, 5218, 6670, 6754], dtype='int64')
```

```
1 df.drop(labels=df[df['tenure'] == 0].index, axis=0, inplace=True)
```

```
2 df[df['tenure'] == 0].index
```

```
Int64Index([], dtype='int64')
```

```
1 df.fillna(df["TotalCharges"].mean())
```

```
1 df["SeniorCitizen"] = df["SeniorCitizen"].map({0: "No", 1: "Yes"})
2 df.head()
```

```
1 def object_to_int(dataframe_series):
2     if dataframe_series.dtype=='object':
3         dataframe_series = LabelEncoder().fit_transform(dataframe_series)
4     return dataframe_series
```

```
1 df = df.apply(lambda x: object_to_int(x))
2 df.to_csv("ibm_telecom.csv")
```

Drop the Customer ID as it is not required



Covert TotalCharges from string to numeric



Drop rows where Tenure is equal to zero



Convert 0 and 1 to Yes and No in Senior Citizen



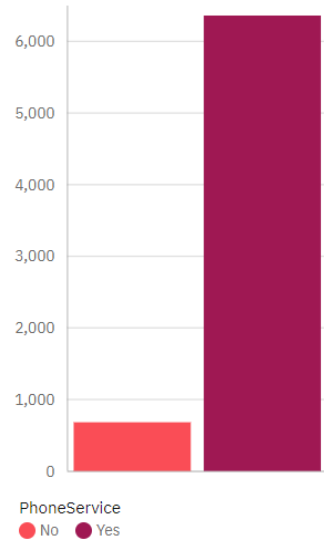
Label Encode all the required column for the further process



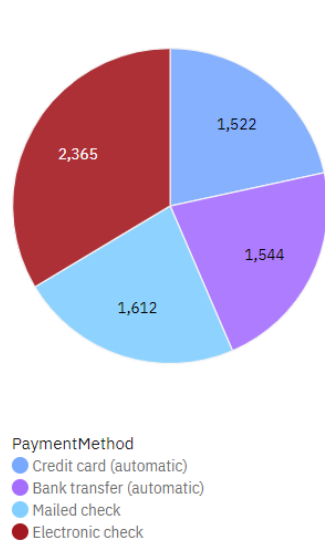
Visualization



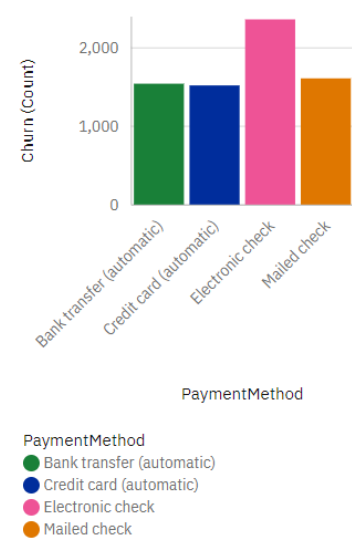
Churn by PhoneService



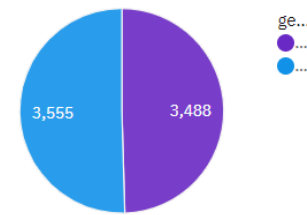
PaymentMethod



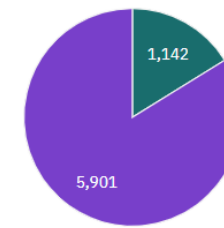
Churn by PaymentMethod



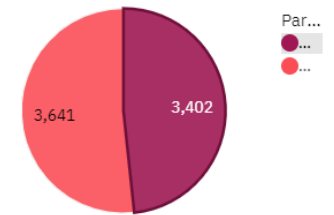
Gender



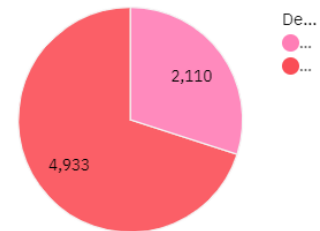
Senior Citizen



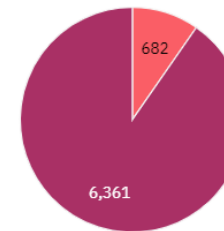
Partner



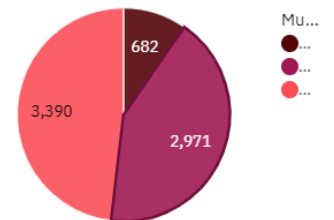
Dependents



PhoneService



MultipleLines



Visualization

Pie Charts

- Distribution of Customers based on various features
 1. Gender
 2. Senior Citizen
 3. Partner
 4. Dependents
- It shows the count of customers in each category

Bar Graphs



Effect of various features on the churn rate



The most important feature affecting the Churn are “Payment methods” and “Phone Service”.

CONCLUSION

- The phase-3 includes us the concept of pre-processing and cleaning of the data.
- The analysis of the Telecom company dataset provided.
- It is done with the help of “IBM –COGNOS” and through the data set which is provided.

T.H.A.N.K.Y.O.U